MIL-S-7804

19 NOVEMBER 1951

Superseding
Navy 94S2(Aer)
16 November 1948

MILITARY SPECIFICATION

SHACKLE, BOMB, MARK 8 MOD 2
MARK 8 MOD 3
MARK 8 MOD 4

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.- This specification covers the manufacture, inspection, and testing of one type of bomb shackle, designated Bomb Shackle, Mark 8, Mod 2, Mod 3, or Mod 4 for use in suspending and releasing bombs, depth charges, etc., from various types of aircraft.

1.2 Classification.- The model designation of the Mark 8 shackle is determined by the steel used for manufacture of the bomb suspension hooks. All other shackle components are identical. The material-s specified for hooks, and the model designation assigned for each material is listed below:

Model	Type Steel Used For Hooks	Applicable Steel Specification
2	AISA 4340	MIL-S-5000
3	AISI 4140	MIL-S-5626
4	HI-TUF	MIL-S-7108

Additional information on hook details is contained in the Aero 1 Type Bomb Shackle Hook Specification MIL-H-6811. (See paragraphs2.1.1 and 2.1.3.)

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of imitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications. -

Military

MIL-H-6811	Hook; Bomb Shackle, Aero I Type
MIL-S-5000	Steel; Chrome-Nickel-Molybdenum (4340) Bars, and Reforging Stock
MIL-S-5626	Steel; Chrome-Molybdenum (4140) Bars, Rods, and Forging Stock (For Aircraft Applications)
MIL-S-7108	Steel, Alloy (Manganese-Silicon-Nickel), Bars and Forging Stock
MIL-T-7743(Aer)	Testing; Store Suspension Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

JAN-P-100	Packaging and Packing for Overseas Shipment - General Specification
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
JAN-P-120	Packaging and Packing for Overseas Shipment - Cartons, Folding, Paperboard
JAN-P-133	Packaging and Packing for Overseas Shipment - Boxes, Set-Up, Paperboard
JAN-P-139	Packaging and Packing for Overseas Shipment - Plywood, Container Grade

Air Force-Navy Aeronautical

AN-P-13	Reservation and Packaging; Parts and Equipment (General Specifications For)

2.1.2 Other Publications. -

Military Standards (Book)

MIL-STD-129 Marking of Shipments

Air Force-Navy Aeronautical Bulletin

No. 43	Specifications and Standards; Use of

2.1.3 Drawings.-

Bureau of Aeronautics Drawings

Sketch 48A4 Bomb Shackle, Mark 8 Mod 2
Sketch 51A144 Bomb Shackle, Mark 8 Mod 3
Sketch 51A145 Bomb Shackle, Mark 8 Mod 4

(Copies of Specifications, standards, and drawings required by contractors in connection with specific procurement fuctions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications shall be of the best commercial quality of the lightest practicable weight, and entirely suitable for the purpose intended.

3.1.1 Critical Materials. - If the materials specified in this specification, applicable applications, and/or drawings for the production of the subject item are critical materials as defined by the National Production Authority, the contractor shall list such materials and notify the procuring agency as to its availability to the contractor. In cases where material is difficult to procure, the contractor shall also prescribe which critical or less critical material can be substituted without departing, revising, or changing the original design and/or function of the subject item. No substitution of such material shall be permitted without prior written approval of the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

3.1.2 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143. except as provided in the following paragraph.

5.1.2.1 AN or JAN Standard Parts.- AN *or* JAN Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by the AN or JAN Standard parts without alteration, and provided the corresponding AN or JAN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN Standard part in effect on date of invitation for bids, commercial parts maybe used provided they conform to all requirements of this specification.

3.2 Purpose.- The bomb shackle covered by this specification shall be capable of suspending and releasing any one U. S. Navy or Amy-Navy bomb or mine of a nominal weight between and including 100 lb and 2,000 lb and equipped with two suspension lugs spaced 14 inches center to center. The shackle is designed for use in both dive and horizontal bombing, and its performance shall comply with the requirements of the most severe applicable conditions listed in Specification MIL-T-7743(Aer), except as otherwise noted therein. The shackle may be actuated electrically by a bomb shackle release or by the manual release system of the airplane, and has provisions for easy latching, and safe locking when suspended or mounted, either externally, as from an airplane wing or fuselage, or internally as within a bomb-bay.

3.3 Interchangeability.- All parts shall be capable of interchangeable assembly, except such parts as are permanently joined to other parts, as by riveting, or welding, in which case the unit thus formed by the permanent joining of parts shall be interchangeable with all other like units. Jigs and gages shall be provided by the contractor for use, where necessary to insure such interchangeability.

3.4 Conformance with Drawings and Specifications.- All component parts shall.be in strict conformity with the applicable approved drawings and material specifications listed in paragraph 2.1.3. No departures from the tolerance shown on the drawings or specifications shall be permitted without prior approval by the procuring agency. Manfacturing processes and finishes shall be as shown unless otherwise approved. (See paragraph 6.1.)

3.5 Workmanship.- The workmanship shall be that required by the best commercial practice covering this type of equipment, for the quantity production of analogous parts. All operating parts of the shackle shall have a smooth workmanlike finish, free from flaws, ragged edges, or tool marks. The exterior shall have a smooth finish with all burrs removed.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of bomb shackles shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the item after the award of contract. to determine that the production meets the requirements of this specification.

(b) Inspection tests: Inspection tests are those tests accomplished on bomb shackles manufactured and submitted for acceptance under contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

4.2 Preproduction Tests.-

4.2.1 Sampling Instructions.- The first five bomb shackles under each contract shall be Preproduction samples shall be delivered for inspection and testing as directed, together with a certification by the cognizant inspector that the materials used in their manufacture are in accordance with pertinent specifications.

4.2.2 Tests.- The Preproduction samples shall be checked for strict conformance with the pertinent drawings and specifications, and for any or all test requirements listed in. Specification MIL-T-7743.

4.2.3 Correction of Defects .- Any defects discovered during the inspection and testing of the Preproduction samples shall be corrected the contractor. Such corrective measures as may be found necessary shall likewise be incorporated in all subsequent bomb shackles offered for acceptance under the contract.

4.3 Inspection Tests. - The bomb shackles, as described and defined herein together with all the materials used in the manufacture thereof, shall be subjected to inspection at the Contractor's plant, or elsewhere, by authorized Government Inspectors.

4.3.1 Materials Inspection-

4.3.1.1 Commercial Materials .- When materials are procured under commercially recognized non-Government specifications, such as by SAE numbers, a certificate showing compliance with such specifications shall be furnished by the supplier of such material. Such certificate shall be in a form acceptable to the Government Inspector.

4.3.l.2 Approval of Materials in Process. - Inspection and approval of material in process during the course of manufacture shall ,not be construed as insuring the acceptance or approval of the finished product.

4.3.2 Individual Tests. - These tests shall be performed by the contractor.

4.3.2.1 Examination of Product.- Each bomb shackle offered for acceptance under the contract shall be inspected to insure that it has been properly assembled and adjusted, and that the workmanship and finish are in keeping with the intent of this specification and applicable drawings in accordance with the requirements of paragraph 3. 5. The parts or dimensions listed below are considered especially critical and shall be given 100 per-cent inspection.

(a) The dimension, 14.187 inches, from the inside edge of one stirrup to the inside edge of the other stirrup as indicated on Drawing 437786, as shown on Sketches 48A4, 51A1144, and 51A145.

(b) Dimension, 1-1/32 inches, from bottom of stirrup pins to top of shackle as indicated on Drawing 437787, as shown. on Sketches 48A4, 51A144, and 51A145.

(c) Positioning of mounting provisions as indicated on Drawings 437786 and 437787, as shown on Sketches 48A4, 51A144, and 51A145.

(d) (d) Mating surfaces between pieces 437790-2 and 437790-7, as shown on Sketches 48A4, 51A144, and 51A145, shall be visually inspected when the shackle is completely assembled for complete area contact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

(e) Hooks shall be checked for requirements in accordance with Specification MIL-H-6811 and Drawing 51A102C1, as shown on Sketches 48A4, 51A144, and 51A145, except that the contractor shall not perform the Fatigue and Ultimate Tensile tests.

(f) Lug bearing and hole dimensions of the hooks shall be checked in accordance with Drawing 51A102Cl, as shown on Sketches 48A4, 51Al44, and 51A145, in order to insure that excessive defamation has not occurred due to prestressing.

(g) Springs shall be tested for compliance with the requirements of the detail drawings.

4.3.2.2 Functional Tests. - Each completely assembled bomb shackle shall be latched (unloaded), a 2-1/4 inch, No. 5 taper pin inserted through the safety pin hole in the sideplates of the shackle, and the release lever of the shackle manually released. The release lever shall return to the latched position by its own spring action, thus relatching the shackle. The taper pin shall be removed, the shackle manually released and the hooks should fully open. The travel and pull of the release lever for 'release,') and the travel of the release lever for "full stroke," shall be measured and shall conform to the "Shackle Release Data" table on Drawing 437786, as shown on Sketches 48A4, 51A144, and 51A145. The above tests shall be repeated with the shackle carrying a 2,000-pound load l

The following number of productions units shall be selected at random from each production lot and subjected to the following tests at the designated Government test activity:

No. of Shackles in Lot	No. of Sample Shackles
99 or less	3
100 to 299	4
300 to 500	

NOTE: Lots over 500 units shall be made up in accordance with the above table.

4.3.3.1.1 Conformance with Drawings - Each sample shall be inspected for strict conformance with the pertinent drawings and specifications.

4.3.3.2 Lag Determination.- All of the samples submitted for each lot shall be submitted to the Lag Determination test, paragraph 4.5, Specification MIL-T-7743.

4.3.3.3 Vibration and Shock Tests. - Half the samples submitted shall be subjected to the Vibration test, paragraph 4.1, and the Shock Test, paragraph 4.8, Specification MIL-T-7743.

4.3.3.4 Life Test and Overload Test.- Half the samples submitted (in case an odd number of samples am submitted subtract one and use one-half the remainder) for each lot shall be subjectd to the Life test, paragraph 4.10, and subsequently to the Overload test, paragraph 4.13, Specification MIL-T-7743. The loads to be used for the Overload test shall be as follows:

21,500 lb Vertically down
12,000 lb Vertically up
9,500 lb Longitudinally forward
9,500 lb Longitudinally aft

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

4.3.4 Rejections.-

4.3.4.1 Sampling Test Samples.- The failure of any Sampling test sample to pass the inspection and tests outlined in paragraph 4.3.2 shall cause tentative rejection of all bomb Shackles represented by that sample, subject to retest as hereinafter provided.

4.3.4.2 Individual Inspection and Test.- The failure of any bomb shackle, any piece, sub-assembly, or assembly of a bomb shackle to pass the inspection and tests outlined in paragraphs 4.3.2.l and 4.3.2.2 above shall cause its tentative rejection subject to retest as hereinafter provided.

4.3.5 Retests.- .

4.3.5.1 Sampling Test Samples - First Retest.- In case any sample representing a production lot fails to pass the tests outlined in paragraph 4.3.2, herein, the contractor shall have the opportunity of reworking the entire affected lot and resubmitting to retest twice as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.3.5.2 Sampling Test Samples - Second Retest.- If, in the Inspectors opinion, the failure of the retest samples is due to a remediable defects the contractor may, at the discretion of the Inspector, be offered the opportunity of reworking the entire affected lot and resubmitting to a second retest three times as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.3.5.3 Individual Inspection and Tests.- Any bomb shackle or any piece, sub-assembly, assembly of a bomb shackle which fails to pass the tests as outlined in paragraphs 4.3.2.1 and 4.3.2.2 may, upon the request of the contractor, be resubmitted for individual inspection and test, provided a statement showing the corrective measures taken, is furnished to the Inspector at the time of such resubmission. Upon retest, failure to meet the requirements outlined in paragraphs 4.3.2.l and 4.3.2.2 shall be cause for final rejection of the lot represented.

4.3.5.4 Salvaged Pieces.- Any piece(s) from a rejected shackle may be salvaged for use providing each salvaged piece is inspected for, and is in complete compliance with, the pertinent drawings and specifications listed on Drawing 48A4.

4.4 Inspection Subsequent to Delivery.- Any or all bomb shackles, subsequent to the first five Preproduction samples, may be subject to inspection and testing for conformance with all requirements and tests of this specification after delivery to the Government.

5. PREPARATION FOR DELIVERY

5. 1 Application.- The preservation, packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation and Packaging.- Each shackle shall be preserved in accordance with the requirements of method 1A, of Specification AN-P-13. No preservation compound shall be required. The units shall be packed in the unlatched condition. Unit containers shall be folding cartons, set-up boxes, or fiberboard boxes, conferring to Specifications JAN-P-120, JAN-P-133, or JAN-P-108.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

5.3 Cushioning. - Unless otherwise specified, cushioning, when used, shall be accomplished with the materials set forth in Specification JAN-P-100. Other nonhygroscopic materials such as wood bracing or blocking, curled hair or latex, or molded papier-mache may be used.

5.4 Packing.- Unless otherwise specified, all shackles shall be afforded domestic packing.
shall be uniform in size, and snugly packed. The gross weight, when packed for shipment, shall not exceed approximately 75 pounds.

5.4.1 Domestic Packing. - Unless otherwise specified, the packages shall be placed in substantial shipping containers, so constructed as to Insure acceptance by common or other carrier, for safe transportation at the lowest rate to the point of delivery specified in the contract or order. Containers shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment, except that fiberboard, when used, shall have a minimum Mullen test of 275 pounds.

5.4.2 Overseas Packing. - Unless otherwise specified, the packages shall be packed in cleated plywood or nailed boxes, conforming to Specifications JAN-P-105 and JAN-P-106, respectively, Plywood, when used, shall be type A or B, condition I, of Specification JAN-P-139. No case liner shall be used.

5.5 Marking and Labeling.-

5.5.1 Packages.- Each unit package shall be durably and legibly marked or labeled with the following information:

BOMB SHACKLE MARK 8 MOD 2, MOD 3,OR MOD 4 (as applicable)
Specification MIL-S-7804
Stock No.
Manufacturer's Serial No.
Contract or Order No.
Quantity
Name of Manufacturer
Name of Contractor (if different from manufacturer)

5.5.2 Shipping Containers.- Each shipping container shall be marked in accordance with the requirements applicable to the individual Services, as specified in Standard MIL-STD-129.

6. NOTES

6.1 Adherence to Drawings and Specifications - Conflict Elimination.- Insofar as the contractor is required by provisions of this specification to adhere rigidly to the drawings listed in Section 2, which may make it impossible in some instances also to comply with the performance and test requirements specified herein, then, in that event, the contractor should promptly notify the procuring agency of such fact, and appropriate changes will be made in the applicable drawings, parts, or performance requirements as authorized by the procuring agency. An equitable adjustment in accordance with applicable contract provisions will be made in the price of the articles to be delivered to reflect any change so authorized by the procuring agency.

6.2 Ordering Data.- Each bid submitted in connection with the procurement of the Bomb-Shackle Mark 8, Mod 2, Mod 3, or Mod 4 shall be accompanied by a statement. showing whether or not any exceptions are being taken to the requirements shown in the drawings or in this specification. If exceptions are taken, they shall be specifically described, and the alternative thereto indicated in the statement.

Source: https://assist.dla.mil -- Downloaded: 20260-17T09:02Z
Check the source to verify that this is the current version before use.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specfications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Navy - Bureau of Aeronautics

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 202610-17T09:02Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL SUPPLY SYSTEMS COMMAND
ENGINEERING DIVISION
ATTN: CODE 0421
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-17T09:02Z
Check the source to verify that this is the current version before use.

MILITARY SPECIFICATION

SHACKLE, BOMB, MARK 8 MOD 2
MARK 8 MOD3
MARK 8 MOD 4

This amendment forms "a part of Military Specification MIL-S-7804 of 19 November 1951, and has been approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

Page 1, paragraph 1.2: Delete and substitute:

"1. 2 Classification - Bomb shackles shall be of the followlng types, as specified(see 6.9).

Type 1 - Mark 8, Mod 2
Type2 - Mark 8, Mod 3
Type 3 - Mark 8, Mod 4

Page 1, paragraph 2.1.1, under Military Specifications: Defense reference to the following:

"MIL-H-6811
MIL-S-5000
MIL-S-5626
MIL-S-7108"

Page 4, paragraph 4. 3.2.1 (a), line 3: Delete "437786" and substitute "48A4D2".

Page 4, paragraph 4. 3.2.1 (c), line 2: Delete "437786" and substitute "48A4D2".

Page 5, paragraph 4.3.2.2, line 0: Delete "437788" and substitute "48A4D2".

Page 5, paragraph 4.3.3.3, line 2: Delete "4.1" and substitute "4.7".

Page 6, paragraph 4.3.5.4: Add

"or Sketch 51A145, as applicable".

Page 7, paragraph 6.2, title: Delete and substitute

'Bid Information"

Page 7: Add as new paragraph 6.3:

Data - Procurement documents should specify the type bomb shackle required (see 1.2)

Custodian:
Navy-Bureau of Aeronautic

FSC 1095

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION
FOR NEW DESIGN

MIL-S-7804
NOTICE 1
17 August 1984

MILITARY SPECIFICATION

SHACKLE, BOMB, MARK 8 MOD 2
MARK 8 MOD 3
MARK 8 MOD 4

This notice should be filed in front of MIL-S-7804, dated 19 November 1951, and Amendment 1, dated 10 December 1957.

Specification MIL-S-7804 and Amendment 1 are inactive for new design and are no longer used by the Departments and Agencies of the Department of Defense except for replacement purposes.

Custodians:

Army - AR
Navy - AS

Preparing activity:

Navy - AS

FSC 1095

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:02Z
Check the source to verify that this is the current version before use.